

SUMMARY:

10+ years of experience in mathematics, computer science, statistics, and machine learning. PhD in operations research; PhD in mathematics; master of quantitative finance. Experienced data scientist. Experienced Python and c++ programmer and financial modeler. Winner of Chinese Mathematical Olympiad.

PROFESSIONAL EXPERIENCE

Bank of America, Quantitative Developer

04/2021-present

- I've proposed and implemented the test plans for commodity, equity, futures, options, fixed income and other financial products hedging strategies
 - Wrote the code files for the hedging strategies in QzDev
 - Tested the codes by writing and running the corresponding unit test files
 - Verified the implemented hedging strategies by capturing the results in SANDRA database and RDS database using SQL server studio and report file in UMA
 - Monitor the performance of the tests on BOB monitor
- Developed the pricing and risk models for interest and interest derivatives products: literature research, data collecting, modeling, and testing on Quartz.
- Designed and implemented a system to compute margins on Quartz.
- Built the Python library including tools to generate tables for interest products from trading data on Quartz.
- Developed the models and the estimation/calibration procedures
- Provided guidance to others through design and code reviews
- Worked with project managers and Risk IT to ensure quant code is seamlessly integrated within the bank's IT systems
- Provided support in development, QA and production issues
- Built a library using python to conduct non-parametric tests on many financial products including credit and its derivative, commodity, index, volatility, bond
- Built a Python library to implement the hybrid VaR method for credit products using Python
 - It combines advantages of both parameterized method and monte carlo method
- Built a library in Python to implement a new machine learning method for the collateral/margin team to identify whether the PV(present value)s of the trades are valid

- Finance team now have fewer (by 6%) false positive to validate, leading to quicker and more accurate responses
- Automation: my predictive model is integrated upstream and escalation is loaded automatically. No manual spreadsheets uploads are required, saving 2+ hours daily
- Reduces disputes among different teams
- Identified the most important predictive features using sk-learn package
- Produced the partial dependence graphs between the target and each feature using pdpbox package

Finra (Financial Industry Regulatory Authority), senior data scientist

09/2020-03/2021

- Implemented a method to detect violations in the option trading **market** using machine learning tools
- Ingested trade blotter data from all the large investment banks.
 - The scope was to normalize and bin data on people's investment accounts - their **risk** tolerance, their options disclosure level approval, **income** levels -- in an effort to identify areas of significant investment loss.
 - Hypothesis was that those exposed - particularly to short-dated options were at elevated **risk** - and we wanted to prove that in the data.
- Parsed a lot of the options information that came in one giant chunk and split them by month of expiration, strike price, ticker symbol, etc.
- Extracted the option trades data from our database using SQL and **Python**, and reorganized them into format that is friendly to business, and saved the results in AWS using Dataiku
- Identified many trading irregularities in option trading. Based on my results, FINRA issued corresponding warning/penalties to many violators.
- Developed a data recording, cleaning and storage mechanism

Wells Fargo, **Quantitative Developer**

01/2020-

08/2020

- Proposed and implemented the outlier detection methodologies using **Python** for various time series with moving average method, density based method like local outlier factor (LOF), and ARIMA method
 - They're successfully applied to interest rate and FX products, resulting in superior results than previous methods
- Wrote the library using **Python** to streamline the reporting process: taking as input the trading files of different format and outputting the summary tables and charts automatically, resulting in great reduction of manual work

- Reduced the trading rejection rate of many products by identifying the factors that impact the rejection rate using machine learning techniques.
 - E.g., for swaption, I found these factors are tenor, strike, expiry, and option types by feature selection

Societe Generale, Quantitative Developer New York
12/2019

06/2018 –

- Proposed and implemented an equity margin method using multi-variant regression with return/volatility against country, industry and market cap, resulting in huge saving of human hours.
- Proposed and implemented a method to detect frauds and irregularities in transactional patterns, using K-mean clustering, feature selection with KNN, and SVM classifier.
- Designed an end-to-end prototype to predict default in order to reduce credit risk, using machine learning tools
- Developed a booking, position keeping and reconciliation mechanism
- Participated in some execution/intra-day alpha research projects
- Shared knowledge and provided technical assistance to other team members within areas of expertise
- Designed and developed the option product's predictive analytics capabilities using disciplined processes, adhering to industry standards and best practices
- Assisted in validating that financial models have been correctly implemented for their intended use
- Designed and developed new software or make modifications to existing one using disciplined processes, adhering to industry standards and best practices, as related to option modeling and all aspects of predictive analytics
- Participated as a team member in the agile process throughout the SDLC to produce high performing, advanced risk analytics for our customers

Voloridge Investment Management, Jupiter, FL, Quantitative Scientist
05/2018

09/2016 –

- I've developed a trading platform that holds the core strategy component, inputs the market data and outputs the trading signals to markets for execution
- I've been researching and implementing performance analytics, including signal performance and post-trade analytics

- Using simple regression and the OU model, I've implemented and backtested a pair trading strategy for stock vs. ETF with JPM and XLF as an example. The result shows it's not very profitable but can be used as a signal input or overlay to another strategy.
- For intraday trading, I've noticed ETH and BTC are loosely related so I proposed, implemented and backtested a StatArb strategy using 90 hour rolling window instead of 90 days for stocks. I'm adding more crypto assets to obtain more insights.
- I've implemented and backtested several strategies based on statistical **analysis**, including channel breakout, dual moving average crossover, momentum, volatility breakout, relative strength index, and moving average convergence/divergence.
- I've improved the software development framework to extend automation and reporting
- I've improved the efficiency and optimization of the analytical library
- I've employed neural networks, decision trees, and deep learning algorithms to generate predictive models that resulted in better strategies:
 - Sharpe ratio: increases from (individual maximum of) 3.38 to 4.51
 - max-drawdown: reduces from 12.4% to 2.79%

Other achievements:

Trading strategy design and backtesting

- Implemented the earning announce premium strategy proposed by Frazzini, using equities to obtain a strategy of Sharpe ratio 0.89
- Investigated the holiday (New Year's Day, Martin Luther King Jr. Day, etc.) effect and implemented a corresponding strategy based on Hansen and Lunder's paper
- Investigated the momentum-factor-and-style-rotation-effect and implemented a corresponding strategy of Sharpe ratio 0.33 based on Tibbs's paper

Multi-factor models **analysis** with **Python** (Pandas, numpy, scikit-learn):

- Used regression tools to predict the price of stocks
 - Linear regression (with regularizations such as ridge, lasso, and elastic net)
 - Regression trees
- Used classification tools to predict the stocks' directions of movement
 - Multi-layer perception (neural network)
 - SVM: linear, polynomial, and Gaussian
 - Logistic regression
 - Tree-based: random forests, boosted-tree (Ada-boost, GBDT, XGBoost)

Natural language **analysis**

- Used NLTK package to analyze the sentiment of news, then predict the price performance of stocks

Deep learning neural network

- Used RNN and LSTM framework to predict the price performance of IBM

Sector clustering **analysis** for equities:

- Proposed connectivity-based algorithm using **Python**, resulting in superior performance in term of stability against adverse volatility movement

Trading decision optimization for equities and futures:

- Used polytopes to approximate ellipsoid feasible region, resulting in much faster speed at the cost of only a little performance (objective **value**)

Pre/post-trade **analysis**

- Used **Python** machine learning tools (scikit-learn) with factors including trading volume and volatility to estimate trading costs.
- Proposed a post-trade **analysis** framework, using Relative Performance Measurement, which overcomes the drawbacks of currently used measurements VWAP and PWP.

Spline polynomial regression:

- Designed and implemented a general framework for spline regression, with the flexibility of arbitrary smoothness and polynomial degree at each knot of Portfolio **analysis**

Scotiabank, Toronto, Canada, **Analyst** (intern)

05/2016 – 08/2016

- Counterparty **risk** measurement using **Python** (Pandas, numpy): PD, LGD, EAD, XVA
- Credit **risk** model development using machine learning method
 - perform feature selection to identify the most suitable features for our binary classification **problem**
 - Chi-squared (p-**value**) test for categorical features
 - ANOVA F-statistic for numerical features
 - calculate the pair-wise correlations of the selected top 20 numerical features to detect any potentially multicollinear variables
 - Weight of Evidence (WoE) and Information **Value** (IV) are used for feature engineering and selection
- **Fixed income modeling** using Hull-White model

Odette School of Business, University of Windsor, Canada, Senior **Research** Scientist

04/2015 –

08/2015

- Disproved a conjecture proposed by Phelim Boyle (pioneer of Monte Carlo methods in finance, inventor of trinomial tree method) on correlation matrices.
- Proposed a better algorithm using **C++** to reduce the effective dimension in Quasi-Monte Carlo method based on linear transformation techniques. The flexibility and superiority of my method is

illustrated by applying it to two high-dimensional applications: Asian basket options and European call options with a stochastic volatility model.

Ivey School of Business, Ontario, Canada, postdoc fellow

2012 - 2015

- Modeled optimal contract problems using principal-agent framework
- Modeled ancillary optimization problems in airlines

Various universities and institutes in China and Canada, visiting researcher

2001-2007

- **Research** in algorithm design and **analysis**

Education

- Master of **quantitative** finance, University of Waterloo
- Ph.D in business, DeGroote School of Business, McMaster University

2017

2013

- Ph.D in Mathematics, Shandong University

2001

- BSc in Mathematics, Shandong University

1996

Awards:

- Top 20 in Chinese Mathematics Olympiad

1992

- Norman Archer Award (Top prize in School of Business, McMaster University)

2010

- Ontario Graduate Scholarship (top graduate award in Ontario, Canada)

2009-2011

Publications:

- 20+ papers with 100+ total citations from top journals in 5+ different areas

Jianbo Qian

- Queens, NY, US

Contact Information

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Summary

Quant with 10+ years of **research** experience in mathematics, computer science, finance, data science and machine learning. PhD in business; PhD in mathematics; master of **quantitative** finance. Experienced **C++ Python** programmer. Winner of Chinese Mathematical Olympiad.

Work History

Total Work Experience: 13 years

- **Quantitative** Developer Bank Of America
Feb 01, 2020
- **Senior Consultant** Societe Generale
Feb 01, 2019
- **Senior Financial Engineer** OCC
Jun 01, 2018
- **Researcher** Voloridge Investment
Sep 01, 2016
- **Analyst** Scotiabank
May 01, 2016
- **Senior Research** Scientist University Of Windsor
Apr 01, 2015

- **Postdoc Fellow Ivey Business School**
Oct 01, 2012

Education

- **Doctorate** | DeGroote School of Business, McMaster University
- **Doctorate** | Shandong University
- **Masters** | University of Waterloo

Languages

- Chinese,

Skills

- **python** - 34 years
- **trading** - 30 years
- **machine learning** - 18 years
- **quantitative analyst** - 16 years
- **research** - 15 years
- **reporting** - 8 years
- **equity** - 7 years
- **monte carlo method** - 7 years
- **finance** - 6 years
- **modeling** - 6 years
- **automation** - 5 years
- **commodities** - 5 years
- **scikit-learn** - 5 years
- **strategy** - 5 years
- **derivatives** - 4 years
- **fixed income** - 4 years
- **pricing** - 4 years
- **quartz** - 4 years
- **algorithm design**
- **anaconda**
- **analysis**
- **analytical skills**

- approximation algorithms
- archer
- black scholes
- business
- c
- canada
- caps
- cds
- china
- cir
- citations
- compact
- conceptual modeling
- credit
- data analysis
- data mining
- design
- efficient
- equities
- equity research
- financial engineering
- finite difference
- floors
- game theory
- graph theory
- health care operations management
- java
- journals
- labels
- languages
- map
- maple
- mathematical finance
- mathematical modeling
- matlab
- member of toastmasters
- mentoring
- microsoft excel
- monte carlo simulation
- network optimization
- networking
- ontario
- operations management
- operations research
- pandas

- principalagent theory
- process scheduler
- public speaking
- publications
- qualitative and quantitative research
- quant
- quantitative analysis
- quantitative analytics
- quantitative data
- quantitative finance
- quantitative investing
- quantitative research
- quantitative risk
- quantitative risk analysis
- r
- rate
- revenue analysis
- revenue management
- risk management
- scheduling
- scholarships
- seaborn
- securities
- short
- simulations
- spss
- statistical data analysis
- statistical modeling
- statistical tools
- statistics
- stochastic calculus
- stochastic methods
- stochastic modeling
- stochastic processes
- stone
- structural engineering
- students
- supply chain management
- swaps
- teaching
- temporary placement
- theoretical computer science
- transportation
- vanilla
- var

- **weight**
- **chinese** - 30 years
- **algorithms** - 15 years
- **time series** - 8 years
- **neural networks** - 3 years
- **bsc** - 1 years
- **qa** - 21 years
- **data science** - 3 years
- **code review** - 2 years
- **computational finance** - 5 years
- **consulting** - 3 years
- **sas** - 5 years
- **value at risk** - 5 years
- **regression analysis** - 4 years
- **credit risk** - 3 years
- **svm** - 3 years
- **cap** - 2 years
- **emc rsa archer** - 1 years
- **investment management** - 1 years
- **mbs** - 1 years
- **pca** - 1 years
- **software implementation** - 1 years
- **treasury** - 1 years
- **software** - 15 years
- **optimization** - 6 years
- **cluster analysis** - 2 years
- **logistic regression** - 2 years
- **bonds** - 1 years
- **payroll** - 1 years
- **science** - 20 years
- **framework** - 6 years
- **ibm** - 1 years
- **infor** - 1 years
- **pd** - 1 years
- **calculus** - 20 years
- **mathematics** - 20 years
- **c++** - 15 years
- **numerical analysis** - 15 years
- **programming** - 15 years
- **xml** - 10 years
- **benefits** - 4 years
- **fx** - 7 years
- **bioinformatics** - 10 years
- **amazon web services** - 6 years
- **database** - 6 years

- **sql** - 6 years
- **mentorship** - 4 years
- **applied mathematics** - 20 years
- **computer science** - 15 years
- **software engineering** - 15 years

Work Preferences

- Desired Work Settings: Remote or On-Site or Hybrid
- Likely to Switch: True
- Willing to Relocate: True
- Travel Preference: 100%
- Preferred Location:
 - FL, US
 - Boston, MA, US
- Work Authorization:
 - US
- Work Documents:
 - Green Card Holder
- Security Clearance: False
- Desired Hourly Rate: 90+ (USD)
- Desired Salary: 150,000+ (USD)
- Third Party: False
- Employment Type:
 - Full-time
 - Contract - Independent
 - Contract - W2
 - Contract to Hire - Independent
 - Contract to Hire - W2

Profile Sources

- linkedin: <https://www.linkedin.com/in/jianbo-qian-2ab36421>
- Dice:
<https://www.dice.com/employer/talent/profile/7880ee5722cf238603ff7de10dbe65b1>